

Sharath Thomas

Senior DevOps Engineer

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PERSONAL PROFILE

Senior DevOps Engineer with 3+ years of experience building cloud-native platforms and CI/CD infrastructure for engineering simulation products. Proven track record of rapid prototyping, cross-team enablement, and turning ambiguous technical challenges into production-ready solutions across Azure, AWS, and Kubernetes environments. Published researcher with an MSc in Advanced Computer Science from the University of Leicester.

EXPERIENCE

Synopsys / Ansys Sheffield, UK

Senior Engineer Apr 2025 — Present

Cross-team platform engineer and go-to contact for integration, deployment, and troubleshooting of Ansys cloud simulation platforms. Led the PoC for containerizing NVIDIA Omniverse Kit Apps (digital twin) and deploying them on OKAS and HPS, building a custom UI for job submission and real-time streaming. Provisioned and managed Azure Kubernetes clusters for OKAS deployments. Contributed to an internal container registry platform with Terraform-provisioned infrastructure for image scanning, promotion, and delivery.

R&D Engineer II Dec 2024 — Apr 2025

Designed and maintained GitHub Actions composite actions and reusable workflows for SAF SDK components and SAF-based applications, automating checks, tests, builds, and release pipelines. Drove the Ansys Fluent webapp streaming PoC enabling browser-based simulation access via HPS and Ansys Cloud Burst Compute. Deployed and debugged AWS Parallel Clusters with Slurm queues for HPS compute scaling. Contributed to developing terraform modules for deploying platform components on single node K3s clusters.

R&D Engineer Jul 2023 — Dec 2024

Automated Ansys HPS deployment on VMs using Ansible, later contributing to the Ansys Simulation Platform initiative as a Docker Compose stack — supporting developers across multiple teams in deployment, maintenance, and troubleshooting. Built a PoC converting SAF platform components (portal API/UI, Keycloak, Nginx, OAuth2-Proxy) to Windows Services for customers unable to use Docker. Managed on-prem and cloud VMs across Azure and AWS for platform hosting and development.

DevOps Intern Jul 2022 — Jun 2023

Created and maintained standardized CI/CD pipelines using GitHub Actions for solution applications. Migrated multiple repositories from Azure DevOps monorepo to GitHub, preserving full git history and pipeline configurations. Built and containerized a React/Flask web application providing UI-based access to internal Ansible automation scripts. Packaged Python developer tools with Nuitka for streamlined distribution.

Samsung R&D Institute India Bengaluru, India

Nov 2019 — Aug 2020

Developed a gesture recognition application for Samsung Watches using Tizen OS, enabling smartwatch-based control of Samsung TVs through intuitive hand gestures for actions like channel switching and volume adjustment. Built sensor data processing pipelines for real-time gesture interpretation on wearable hardware.

ISA-VIT (International Society of Automation) Vellore, India

Dec 2017 — Jun 2020

Progressed from Core Member to Technical Head of the student chapter. Conducted weekly "Technitudes" sessions introducing students to emerging technologies in automation, IoT, and software development. Co-ordinated an international research conference for academic paper publications as part of the board. Led and mentored 100+ members, organising hands-on workshops and events that made the chapter a go-to community for skill development on campus.

EDGATE Technologies Bangalore, India

Jun 2019 — Jul 2019

Embedded Systems and IoT internship focusing on hardware-software integration using Texas Instruments Launchpads.

EDUCATION

MSc Advanced Computer Science with Industry Sep 2021 — Sep 2023

University of Leicester

Passed with Merit. Course topics include Cloud Computing and Mobile and Web Applications.

BTech Electronics and Communications Engineering with Specialization in IoT and Sensors Jul 2017 — Jul 2021

VIT Vellore

CGPA: 9.02/10. Course modules include DSA, Java, C++, Python, Web Technologies and Mobile App Development.

SKILLS

CI/CD & Automation: GitHub Actions, Azure DevOps Pipelines, Tekton

Containers & Orchestration: Docker, Docker Compose, Kubernetes, K3s, Helm

Cloud Platforms: Azure, AWS, GCP

Infrastructure as Code: Terraform, Ansible

Platform & Networking: Traefik, Nginx, Keycloak, OAuth2-Proxy, Slurm

Languages: Python, Go, JavaScript, TypeScript, Bash, HCL, Java

Frontend & Backend: React, FastAPI, Node.js, PostgreSQL

DevOps Tooling: Git, Linux, Nuitka, Zot (OCI Registry)

CERTIFICATIONS

- Kubernetes and Cloud Native Associate (KCNA)
- Architecting with Google Kubernetes Engine Specialization
- Architecting with Google Compute Engine Specialization
- Industrial IoT on Google Cloud Platform
- Java for Android